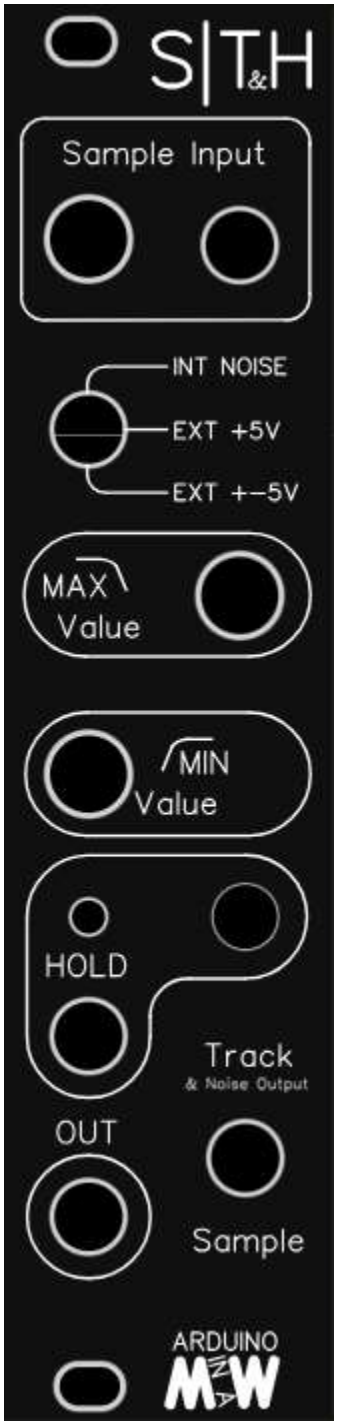
Users manual for the MiaW S|TH



The MiaW SITH is a Sample (or Track) and Hold with a twist built around an Arduino. The

schematics and code is open source for personal use and can be downloaded and modified from the github page of the product. It can also be bought as a PCB+Panel set or as a full kit from this store:

In its basic form it is a Sample & Hold, but can also be a Track and Hold, and a Noise source. It then also has a bunch of alternate functions which gives it a few more ways to work.

The controls

- the Sample input with an attenuverter

- a switch to chose either external samples in either 0.5v or -5 to +5v or internal noise. For noise output set this knob to internal noise.

- max and min knobs, in S&H these will constrain the sample material within the min and max values to make the output signal more collected.

- Hold button and gate input, for holding the signal.

- a switch to set either Sample or Track and hold. For noise output set this switch to Track

Function

In S&H, Each time you press the Hold button or send a gate to the hold input a sample will be made of the sample material and held for as long as the button or gate is held. The sample material can be anything, but the output will always be a CV signal.

In T&H, the input signal will be on the output while the Hold is not activated. Once it is activated it will again sample and hold the signal at that moment.

For noise output set the sample input to internal noise, and the Track or Sample switch to Track. This will result in noise on the output.

In any of these functions turning the max and min knobs will restrict the range of the CV signal on the output to stay within those two knobs, putting them very close to each other will result in a very close set of CV signals.

Alternate functions

To reach the alternate functions you need to set both MAX and MIN knobs furthest ccw, and press the HOLD button. For each two seconds the Hold LED will either turn on or off, each time it changes is a change of function. The function you choose will be saved to memory and used when the module is power cycled or until changed with the same procedure.

The functions are as follows:

1. Default function Sample (or Track) and Hold
2. Random gates - Bernoulli gate with only one output. When a gate or trig is recieved on the Hold input, the MAX value is used to decide if it is routed to the output or not. at CCW all gates go through and at full CW no gates go through.
3. Sample & Slide - In this mode the Max knob decides how fast the cv value should slide down and MIN how fast it slides upwards between the last sampled value and the next. When the knobs are at ccw no slide occurs, and it acts just as a S&H, at full cw it slides 100% of the time between the samples.
4. LFO & Hold - In this mode the MIN and MAX creates a saw/triangle/ramp wave where MIN sets the rising edge, and MAX sets the falling edge. In Track mode you will get an LFO out, and in Sample mode you will get an LFO you can sample from. You can also sample in Track mode...
5. AD envelope - This is a simple AD envelope based on the code from the ADSR but with much less resolution. Very similar to the LFO & Hold but the envelope is triggered by a gate or trig on the HOLD input. The MIN sets the Attack and MAX sets the DECAY.
6. Percussive Noise - This can be used for snares and such, where the internal noise is activated when the HOLD is activated. The MIN knob sets the Decay of the noise and MAX sets sample reduction for some really lofi noise bursts. **This can also double as a LoFi VCA** if using external sound instead of internal noise.